

Poster: GEM-PLI Side Channel in GPON Inferring Encrypted Video Streaming from Access-Layer Length Traces

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Abstract—Over-the-top (OTT) video usage continues to grow, making *video identification attacks*—inferring what a user watches from encrypted traffic patterns—a concrete privacy threat. GPON encrypts downstream GEM payloads, yet it must expose access-layer framing metadata in plaintext. This poster provides early evidence that the GEM Payload Length Indicator (PLI) can act as a privacy side channel: a co-located observer under the same optical splitter can record a high-rate time series of encrypted payload lengths (no AES decryption) that preserves HTTP adaptive streaming (HAS) burst structure. We build a GPON downstream trace generator that maps captured encrypted YouTube/Netflix sessions into GEM units and packs them into 125 μ s downstream frames, enabling controlled evaluation. Using only PLI-derived signals (PLI sequences or per-frame occupancy), we show preliminary separability between streaming and non-streaming and provide indicators of adaptive bitrate (ABR) transitions. We discuss mitigation directions (padding/shaping and privacy-aware handling of access-layer metadata).

I. INTRODUCTION

End-to-end encryption (HTTPS) protects video content, but it does not eliminate side channels. Video identification attacks exploit the structure of HTTP adaptive streaming (HAS): videos are delivered as short segments, and the resulting size/timing bursts depend on the representation (codec/bitrate) and content. Most prior work observes traffic at the IP/Ethernet layer, where segment patterns are blurred by transport and host effects (TCP segmentation, retransmissions, congestion control, and OS scheduling), often requiring longer windows or heavier training.

Why GPON may be “cleaner.” GPON introduces an underexplored vantage point *below* IP: GEM headers must remain plaintext for real-time multiplexing, and they expose the Payload Length Indicator (PLI). Combined with GPON’s fixed 125 μ s downstream framing, this yields a scheduling-aligned, high-rate length signal that is less directly affected by host-side packetization artifacts than IP-layer packet sizes.

Takeaways (this poster). (i) We demonstrate a GPON trace generator that produces access-layer observables

(PLI/occupancy) from legitimate encrypted video traffic; (ii) we provide preliminary evidence that PLI-derived signals can detect streaming sessions and indicate ABR changes in a controlled test OLT testbed; (iii) we outline mitigation directions and an evaluation method to quantify privacy–overhead trade-offs.

II. THREAT MODEL

We consider a metadata-only attacker co-located under the same optical splitter as a victim subscriber.

- **Capability:** passively records downstream GEM headers (no payload decryption; no OLT/operator access).
- **Observable:** a length time series from GEM-PLI (or per-125 μ s frame occupancy) for the victim’s Port-ID.
- **Goals:** (1) streaming vs. non-streaming, (2) ABR quality-transition detection, and (3) (future work) content fingerprinting given a reference catalog.
- **Ethics:** all measurements use the author’s own traffic on a controlled testbed; we analyze metadata only.

III. TRACE GENERATION (125 μ s)

To avoid relying on proprietary GPON capture hardware, we build a GPON downstream trace generator. From a PCAP of encrypted video traffic, we (i) map Ethernet frames into GEM payloads (fragmenting to satisfy the 12-bit PLI limit), (ii) generate plaintext GEM headers (PLI/Port-ID/PTI/HEC), (iii) encrypt payload bytes with AES-CTR (header remains plaintext), and (iv) pack the GEM stream into fixed-length 125 μ s downstream frames consistent with the configured line rate. The attacker signal is then either the GEM-level PLI sequence or the per-frame occupancy series.

IV. PRELIMINARY RESULTS

In controlled sessions, PLI-derived signals exhibit segment-driven burst structure consistent with HAS. Using simple windowed features (e.g., sums/variance/length histograms) we observe preliminary separability between streaming and non-streaming. We also see distribution shifts consistent with ABR changes, motivating a lightweight change-point detector and, longer-term, learned classifiers.

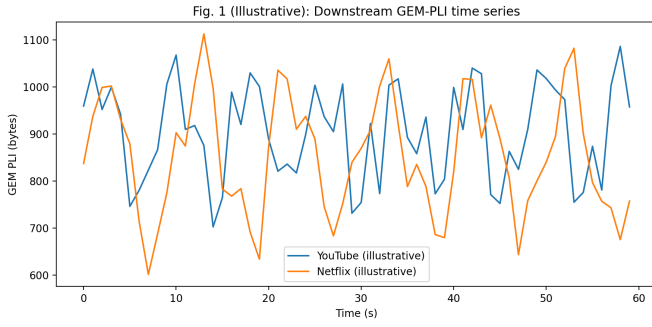


Fig. 1. **Fig. 3 (one-line).** Example GEM-PLI (or 125 μ s frame occupancy) time series for YouTube vs. Netflix sessions, highlighting segment-driven burst structure and candidate ABR transition points.

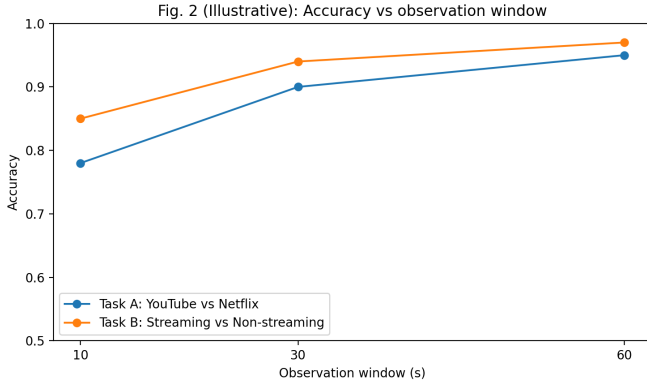


Fig. 2. **Fig. 4 (one-line).** Classification accuracy vs. observation window length (e.g., 10/30/60 s) for streaming-session detection and preliminary ABR-change detection.

a) Validation protocol.: To validate that these results are not overfitting or dataset leakage, we will (i) use session-level splits (train/test split by video/session, not by window) to prevent overlapping segments across splits, (ii) report confidence intervals via bootstrapping over sessions, (iii) include baselines (random guess, IP-layer-only features, and a simple threshold detector) to contextualize gains, and (iv) perform ablations over background traffic and multiplexing intensity to show whether the signal persists when realistic noise is introduced.

V. IMPACT & MITIGATION

Impact. If GEM-PLI leaks stable HAS structure, payload encryption alone may be insufficient to protect behavioral privacy in optical access networks. The leakage can enable inferences ranging from “is streaming occurring?” to quality profiling (ABR transitions), and potentially content fingerprinting with a larger reference dataset.

Mitigation directions.

- *Padding/shaping:* flatten GEM-PLI or per-frame occupancy dynamics at the cost of bandwidth overhead.
- *Privacy-aware access-layer design:* quantize/aggregate length metadata or redesign framing to reduce fine-grained leakage.

REFERENCES

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